

```
-- 1. Retrieve all books from fiction genre.
```

```
SELECT
```

```
    *
```

```
FROM
```

```
    books
```

```
WHERE
```

```
    genre = 'fiction';
```

-- 2. Find books published after the year 1950

SELECT

*

FROM

books

WHERE

Published_Year >1950;

```
-- 3. List Customers from Canada
```

```
SELECT
```

```
    *
```

```
FROM
```

```
    customers
```

```
WHERE
```

```
    country="canada";
```

-- 4. Orders Placed in november 2023.

SELECT

*

FROM

orders

WHERE order_date between "2023-11-01" and "2023-11-30";

```
-- 5. total books available  
SELECT  
    SUM(stock) AS total_count  
FROM  
    books;
```

```
-- 6. find details of most expensive books
```

```
SELECT
```

```
    *
```

```
FROM
```

```
    books
```

```
ORDER BY price DESC
```

```
LIMIT 1;
```

```
-- 7. Show all customers who ordered more than 1 quantity of books
SELECT
    *
FROM
    customers c
    JOIN
    orders o ON c.customer_id = o.customer_id
WHERE
    o.quantity > 1;
```

```
-- 8. retrieve all orders where total amount exceeds $20
```

```
SELECT
```

```
    * from
```

```
    orders where total_amount > 20;
```


-- 9. List all genre available in the book table

SELECT

 distinct genre from books;

-- 10. find the book with the lowest stock

```
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```

```
SELECT
```

```
    *
```

```
FROM
```

```
    books order by stock asc limit 1;
```

-- 11.Calculate the total revenue generate from the tables

```
select * from orders;
```

```
select sum(total_amount) as total_rev from orders;
```

-- 12. Retrive the total number of books sold for each genre

SELECT

b.genre, sum(o.quantity) as total_books

FROM

books b

JOIN

orders o ON b.book_id = o.book_id group by genre;

```
-- 13. Find the average price of books in the fantasy genre.  
select avg(price) as average_price from books where genre="fantasy";
```

```
-- 14. List the customers who have placed atleast 2 orders
SELECT
    c.name, COUNT(o.order_id) AS order_count
FROM
    customers c
    JOIN
    orders o ON c.customer_id = o.customer_id
GROUP BY c.name
HAVING COUNT(o.order_id) >= 2;
```

```
-- 15. find the most frequently ordered book  
  
SELECT  
    b.title, COUNT(o.order_id)  
  
FROM  
    books b  
    JOIN  
        orders o ON b.book_id = o.book_id  
  
GROUP BY b.title  
  
ORDER BY COUNT(o.order_id) DESC limit 1;
```

-- 16 Show the top 3 most expensive books of "fantasy" genre :

SELECT

 b.title, b.price

FROM

 books b

WHERE

 genre = 'fantasy'

ORDER BY price DESC

LIMIT 3;


```
-- 17 Retricve the Total Quantity of books sold by each author
SELECT
    b.author, SUM(o.quantity) total_quantity
FROM
    books b
    JOIN
        orders o ON b.book_id = o.book_id
GROUP BY b.author
ORDER BY total_quantity DESC;
```

```
-- 18 list the cities where customers who spent over $30 are located:
SELECT
    c.city, o.total_amount
FROM
    customers c
    JOIN
    orders o ON c.customer_id = o.customer_id
WHERE
    total_amount > 30
ORDER BY total_amount DESC;
```

-- 19.find the customer who spend more on orders :

SELECT

c.customer_id, c.name, SUM(o.total_amount) AS Total_spent

FROM

customers c

JOIN

orders o ON c.customer_id = o.customer_id

GROUP BY c.customer_id

ORDER BY total_spent DESC limit 1;

-- 20 Calculate the stock remaining after fulfillment of all the orders :

```
SELECT
    b.book_id,
    b.title,
    b.stock,
    COALESCE(SUM(o.quantity), 0) AS total_quantity,  b.stock-COALESCE(SUM(o.quantity), 0) as remaining_quantity
FROM
    books b
    LEFT JOIN
    orders o ON b.book_id = o.book_id
GROUP BY b.book_id;
```